Dual-Expert Keys for Reducing False-Positive Retrieval in Lifelong Vision–Language Model Editing

Anonymous submission

Abstract

Lifelong vision–language model editors store sequential edits in a growing expert pool and retrieve relevant experts at inference time. Although these systems can achieve high reliability on edited examples, routing logs reveal a distinct grounding failure: when the prompt is held fixed but the image is swapped or replaced by a hard negative, the editor frequently activates an expert associated with a different scene. Standard reliability, generality, and locality metrics do not expose this behavior. We formalize it as false-positive retrieval and evaluate it using adversarial grounding probes constructed from VLKEB and EVQA. To reduce this failure, we introduce Dual-Expert Keys (DEK), which allow an expert to activate only when it passes both a prototype-relative visual gate and a prompt-key gate. We further introduce adversarial visual binding training (AVBT), which strengthens edit–image association by suppressing expert activation on mismatched images. On a BLIP-2 OPT-2.7B lifelong mixture-of-experts editor, DEK reduces VLKEB image-swap false-positive retrieval from 96% to 46% across five evaluation seeds while preserving 93.7% reliability and maintaining standard generality and locality. Additionally, on a cross-dataset setup on EVQA using hard-negative images, false-positive retrieval fall from 96.8% to 2.7%, with reliability remaining near 95%. These results identify retrieval leakage as a distinct failure mode in lifelong multimodal editing and demonstrate that dual-modal gating can substantially improve retrieval safety without sacrificing edit reliability.

Introduction

Large vision-language models (VLMs) can answer questions about photographs, documents, and everyday scenes, and increasingly serve as implicit knowledge bases in interactive systems (Li et al. 2023). However, the knowledge encoded in these models may become outdated or inaccurate as entities change, annotations are corrected, and new visual information emerges after pretraining. Updating the entire model through fine-tuning is computationally expensive and may degrade orthogonal capabilities. Model editing therefore provides an attractive alternative by introducing targeted corrections while preserving model behaviour on unrelated inputs (Mitchell et al. 2021, 2022; Cheng et al. 2023). Multimodal editing benchmarks such as MMEdit and VLKEB evaluate this objective through reliability, generality, and locality (Cheng et al. 2023; Huang et al. 2024). In lifelong editing, the challenge is further complicated by a sequence of updates that must coexist without overwriting previously acquired knowledge (Chen et al. 2025).

Retrieval-based and mixture-of-experts editors are well suited to lifelong editing because they can store each update as an expert or memory entry while leaving the backbone model unchanged (Mitchell et al. 2022; Chen et al. 2025). At inference time, the editor retrieves the experts considered relevant to the current query and injects their corresponding residual updates. Although these systems can achieve high reliability on the images used during editing, high reliability does not necessarily imply that the edited knowledge is visually grounded. For example, consider an edit that associates a photograph with the answer "Kankakee" (Figure 1) for the question "What city in Illinois is depicted in the image?" The edited model may answer correctly when presented with the original photograph, yet continue to produce the same answer when the photograph is replaced by an unrelated skyline or a visually confusable scene. This problem becomes increasingly consequential as the expert pool grows because additional stored edits create more opportunities for an unrelated expert to be selected.

Our analysis indicates that this behaviour is frequently caused by incorrect expert selection rather than interference with the parameters of the frozen VLM. On the proposed adversarial grounding probes, routing logs show that experts associated with other images are often activated at rates exceeding 90%. We refer to this failure as *false-positive retrieval*. It occurs when an unrelated stored expert receives a sufficiently high retrieval score to pass the routing threshold and inject an incorrect fact. False-positive retrieval is therefore a routing safety failure that is distinct from answer-level accuracy. Standard evaluations based on reliability, generality, and locality do not reveal which expert is activated when the prompt is preserved but the image is replaced by a mismatched or confusable alternative (Cheng et al. 2023; Huang et al. 2024; Chen et al. 2025).

Existing multimodal editors do not directly address this failure. LiveEdit combines visual filtering with textual routing, but its retrieval mechanism and evaluation are designed

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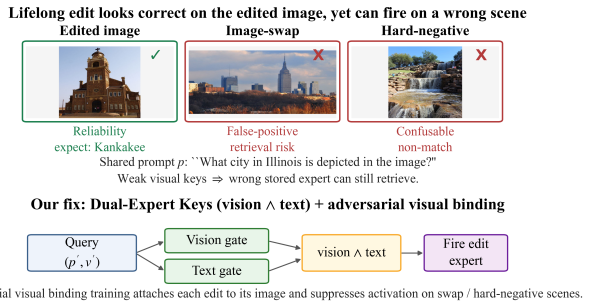


Figure 1: Top: shared prompt with the edited image (reliability), an image-swap, and a hard-negative. Bottom: Dual-Expert Keys fire an edit only when vision and text gates both pass; adversarial visual binding strengthens edit–image attachment under these probes.

primarily around reliability, generality, and locality rather than the activation of unrelated experts under adversarial images (Chen et al. 2025). DualEdit investigates the placement of modality-specific adapters and the resulting trade-off between reliability and locality, rather than retrieval leakage in a growing expert pool (Shi et al. 2025). Other editing approaches, including Fine-tuning, MEND-VL, and SERAC-VL, provide relevant comparisons for answer-level editing performance but do not expose the same growing set of expert identities required to measure false-positive retrieval directly (Mitchell et al. 2021, 2022; Cheng et al. 2023). Consequently, a system may perform well under standard editing metrics while still activating an unrelated stored edit when the visual context changes.

Addressing this problem requires modifying both expert retrieval and the association between an edit and its source image. Strengthening only the visual binding of an expert is insufficient when unrelated experts can still pass the retrieval threshold. Similarly, routing based on either visual or textual evidence alone can admit incorrect experts. A visual gate may accept an expert with a colliding image representation, whereas a textual gate may accept an expert because multiple edits share similar prompts. We therefore require agreement between the visual evidence and the prompt before an expert can be activated.

Based on this principle, we introduce DEK, a retrieval mechanism that maintains separate visual and textual keys for each stored edit. At inference time, an expert is activated only when it satisfies both a prototype-relative visual gate and a prompt-key gate. We complement this mechanism with AVBT, which encourages activation on the correct edited image while suppressing expert activation on image swaps and hard negatives. DEK directly changes the routing decision, whereas AVBT strengthens the visual association on which the routing mechanism operates. We instantiate both components within a lifelong mixture-of-experts editor using BLIP-2 as the frozen VLM backbone (Li et al. 2023; Chen et al. 2025).

Experiments on adversarial grounding probes constructed from VLKEB demonstrate that DEK reduces the image-swap false-positive retrieval rate from 96% to 46% across five evaluation seeds while preserving 94% reliability. Reliability, textual generality, multimodal generality, and locality remain comparable to those of the lifelong baseline. Component ablations show that adversarial binding alone leaves the false-positive retrieval rate near 96%, indicating that the dual-expert retrieval mechanism is responsible for the substantial reduction in incorrect expert activation. The same pattern generalizes to EVQA, where DEK reduces hard-negative false-positive retrieval from 96.8% to 2.7% while preserving reliability near 95%. Comparisons with Fine-tuning, MEND-VL, and SERAC-VL on the same adversarial probes further show that DEK uniquely maintains high reliability while directly reducing incorrect expert retrieval. Our objective is to prevent retrieval leakage rather than to achieve state-of-the-art token-level accuracy under image swaps.

The main contributions of this work are as follows:

- We identify and formalize false-positive retrieval as a visual grounding failure in lifelong multimodal editing. We introduce image-swap and hard-negative probes that expose incorrect expert activation not captured by standard reliability, generality, and locality metrics.
- We propose Dual-Expert Keys, which require agreement between visual and textual retrieval signals before an expert can be activated. We further introduce adversarial visual binding training to strengthen the association between each edit and its corresponding image.
- We demonstrate substantial and stable reductions in false-positive retrieval in both within-benchmark on VLKEB and cross-dataset setup on EVQA. On both datasets, DEK reduces false-positive retrieval while preserving high reliability and standard editing performance.

Related Work

Knowledge editing for language models. Modern language models store large amounts of factual knowledge, but that knowledge can be wrong, outdated, or incomplete. Model editing aims to inject a local correction without full retraining and without broadly damaging unrelated behaviour (De Cao, Aziz, and Titov 2021; Yao et al. 2023). The literature is usually grouped into three families. Locate-then-edit methods first identify parameters associated with a fact and then rewrite them; ROME does this for single edits, and MEMIT scales the idea to many facts (Meng et al. 2022a,b). Meta-learning editors such as MEND train a hypernetwork to turn edit gradients into low-rank weight updates (Mitchell et al. 2021). Memory-based retrieval editors keep the base model frozen and store edits externally. SERAC retrieves a counterfactual memory and routes through a scope classifier (Mitchell et al. 2022), GRACE maintains lifelong discrete key-value adaptors (Hartvigsen et al. 2023), and in-context editors such as IKE place edit demonstrations in the prompt (Zheng et al. 2023). EasyEdit unifies many of these methods behind a common interface, which has made fair comparison easier (Wang et al. 2024). Beyond single-fact reliability, later evaluation work asks whether an edit propagates consistently to related facts (Cohen et al. 2024) and whether editors can handle long or diverse-format targets (Jiang et al. 2025a). AnyEdit extends long-form text

editing; it decomposes targets into chunks and updates them autoregressively, one segment at a time rather than a single pass (Jiang et al. 2025a).

Multimodal knowledge editing and benchmarks. Editing vision-language models is harder than editing text-only language models because the fact is tied to an image as well as a prompt (Cheng et al. 2023). MMEdit established early multimodal reliability, generality, and locality protocols for VQA- and caption-style edits (Cheng et al. 2023). VLKEB improves visual quality by building edits from multimodal knowledge graphs and adds portability-oriented checks (Huang et al. 2024). More recent benchmarks push different axes. MC-MKE emphasizes modality consistency across fine-grained edit formats (Zhang et al. 2025), MMKE-Bench covers diverse visual semantics and free-form knowledge (Jiang et al. 2025b), and ComprehendEdit aggregates multiple tasks with generalization and preservation indices (Ma et al. 2025). On the method side, DualEdit studies where to edit. It updates text and vision at modality-sensitive layers and uses a gate to protect original behaviour (Shi et al. 2025). That gating is about reliability-locality trade-offs under layer choice, not about blocking wrong-expert retrieval in a lifelong MoE pool. Across this literature, success is still mostly reported as whether the new answer appears on the edited (or rephrased) input, and unrelated inputs stay stable. What is usually missing is an explicit account of which stored edit fired when the prompt is held fixed and the image is swapped or made confusable.

Lifelong retrieval editors and visual keys. Lifelong editing requires many updates to coexist over time (Hartvigsen et al. 2023; Chen et al. 2025). LiveEdit is the closest system to our setting. It generates a low-rank MoE expert per edit, stores experts in a growing repository, and at inference uses hard visual filtering followed by soft text routing to fuse relevant experts into a frozen VLM such as BLIP-2 (Chen et al. 2025; Li et al. 2023). Related multimodal backbones used in editing studies include LLaVA and InstructBLIP (Liu et al. 2023; Dai et al. 2023). Because retrieval decides which residual is injected, routing quality becomes part of editing safety as the pool grows an issue that sparse MoE selection has long faced in other domains (Shazeer et al. 2017). Strong visual encoders can help in principle. CLIP-style vision-language similarity (Radford et al. 2021) and DINOv2 features (Oquab et al. 2024) often separate images well in offline probes. But offline discrimination does not automatically transfer to edit retrieval when the editor compresses vision into a few routing keys and scores them against a large lifelong pool. Weight-updating multimodal adaptations of Fine-tuning, MEND, and SERAC remain important baselines (Mitchell et al. 2021, 2022; Cheng et al. 2023); they can change answers, yet they do not expose a growing expert identity that we can score for false-cross.

Taken together, prior editing methods and multimodal benchmarks have made reliability, generality, and locality the default success criteria (Cheng et al. 2023; Huang et al. 2024; Chen et al. 2025). Those criteria ask whether the edited answer appears on the intended or rephrased input, and whether unrelated inputs remain stable. They do not ask, under a shared prompt and an adversarial image, whether a lifelong pool retrieves a foreign expert. As a result, false-positive retrieval under image-swap and hard-negative conditions remains largely unmeasured, even in settings where continuous expert pools make that failure observable.

Problem Defination

Lifelong multimodal editors that store one expert per edit are attractive precisely because the base VLM can stay frozen. As edits accumulate, however, the system is only as safe as its retrieval step. Write $\mathcal{E} = \{e_1, \ldots, e_m\}$ for the current expert pool, and let each e_i remember an edit (p_i, x_i, y_i) a prompt, an image, and a target answer. Given a new query (p, x), the editor scores the pool and injects one or more residuals into the frozen model. On the photograph that was actually edited, this design can look excellent: reliability, generality, and locality numbers in the literature are often already high (Cheng et al. 2023; Huang et al. 2024; Chen et al. 2025). The awkward case is the one those scores largely ignore. Keep the prompt fixed, replace the image with something else, and ask: does the pool still hand you an expert that belongs to a different scene?

To make that question concrete we use a small suite of *adversarial grounding* probes built around the same edit prompt p_i whenever we can. Reliability puts the edited image x_i back in front of the model; it answers the usual question of whether the edit took. *Image-swap* instead feeds another edit's image x_j ($j \neq i$), so the text still asks for y_i while the pixels show a different entity or place. *Hard-negative* keeps p_i but substitutes a near-neighbour distractor that was never itself an edit image. Text rephrases and ordinary locality checks stay in the protocol as controls. If a text rephrase roughly tracks reliability, yet swap and hard-negative collapse, the trouble is unlikely to be "the language model simply ignored the picture." Along the way we record not only token overlap with y_i, but also which pool experts actually fired routing identities and how many were retrieved.

That logging lets us separate two notions that are easy to conflate. Answer-level swap or hard-negative accuracy tells you what the model said. We care at least as much about which memory was opened. We call it *false-positive retrieval* when, for an adversarial query aimed at edit i, retrieval activates at least one expert e_k with $k \neq i$. The rate of that event is our main leakage number (lower is better). In tables we write it **FPR**. FPR is deliberately a retrieval-safety metric. A non-zero token score on a wrong image can happen for many reasons, including ordinary generation from the frozen backbone. A high FPR, by contrast, points at the lifelong pool: the editor selected a foreign expert and injected the corresponding fact.

Standard Rel, Gen and Loc reporting does not catch this. Those metrics never ask, under a shared prompt and an adversarial image, whether the retrieved expert identity matches the scene on screen. So a system can post strong on these while still admitting colliding visual keys as the pool grows. Figure 1 is the picture we have in mind: one prompt, three images (edited, swapped, hard-negative), and later a dual vision-text check before an expert is allowed to fire.

The empirical picture on our BLIP-2 lifelong MoE testbed (Li et al. 2023; Chen et al. 2025) with VLKEB-derived probes (Huang et al. 2024) is hard to dismiss as a soft-overwrite story alone. On the VLKEB evaluation split with our adversarial probes, reliability for the lifelong baseline sits near 94%, yet swap FPR climbs to 96%: the edit looks solid on its own photograph and still almost always retrieves someone else's expert under image-swap. More broadly we see the same shape again and again strong own-image reliability, a wide gap once the image is swapped or made confusable, and FPR often above 90% when many experts share a scalar visual key. Results will show that teaching the editor to bind more tightly to its own image is not enough by itself. If foreign experts keep clearing the retrieval bar, the decision rule has to change: an edit should fire only when the image evidence and the prompt key both agree. That is the motivation for Dual-Expert Keys at inference and for adversarial visual binding during training.

Methodology

We target false-positive retrieval directly with two coupled mechanisms (Figure 2). *Dual-Expert Keys* change the inference rule: a stored edit may fire only when its vision evidence and its prompt-key evidence both agree with the query. *Adversarial visual binding* is the training counterpart: each edit is attached to its correct image and suppressed on swap and hard-negative images, so the dual gate is not fighting weakly grounded keys. The two play different roles empirically binding improves adversarial answer behaviour somewhat, while Dual-Expert Keys are what cut false-positive retrieval roughly in half. We instantiate the method on a frozen VLM with a growing pool of low-rank residuals, because that setting makes wrong-expert retrieval observable.

Expert pool and keys

Let the frozen VLM be f_θ. An edit (p_i, x_i, y_i) produces mid-layer representations from which we build: (i) an expert e_i (low-rank MoE residual parameters); (ii) a Q-Former visual key $\mathbf{k}_i^{\mathrm{v}}$; (iii) a patch-grounded visual key $\mathbf{k}_i^{\mathrm{p}}$ from a DINOv2 patch grid passed through a small PatchGroundingModule; (iv) a prompt key $\mathbf{k}_i^{\mathrm{t}}$. After t edits the pool is

$$\mathcal{E}_t = \left\{ \left(e_i, \mathbf{k}_i^{\mathrm{v}}, \mathbf{k}_i^{\mathrm{p}}, \mathbf{k}_i^{\mathrm{t}}\right) \right\}_{i=1}^{t}. \tag{1}$$

On a query (p', x') we extract query keys $(\mathbf{q}^{\mathrm{v}}, \mathbf{q}^{\mathrm{p}}, \mathbf{q}^{\mathrm{t}})$ in the same spaces. Throughout, $\mathrm{sim}(\mathbf{u}, \mathbf{v})$ denotes scaled cosine similarity between ℓ_2-normalized vectors (equivalently a temperature-scaled dot product); the learned scale is absorbed into sim and is shared across vision and text routing. Vision logits mix the Q-Former and patch scores (with a fixed fuse weight $\alpha \in [0, 1]$ when both are present):

$$s_i^{\mathrm{v}} = \alpha \,\mathrm{sim}\big(\mathbf{q}^{\mathrm{v}}, \mathbf{k}_i^{\mathrm{v}}\big) + (1 - \alpha)\,\mathrm{sim}\big(\mathbf{q}^{\mathrm{p}}, \mathbf{k}_i^{\mathrm{p}}\big), \tag{2}$$

and likewise a text logit $s_i^{\mathrm{t}} = \mathrm{sim}(\mathbf{q}^{\mathrm{t}}, \mathbf{k}_i^{\mathrm{t}})$. A non-edit "prototype" vision score $s_{\mathrm{prot}}^{\mathrm{v}}$ is computed from the query itself and acts as the no-edit baseline for the vision gate.

Dual-Expert Keys

False-positive retrieval is, in our logs, mostly a *retrieval agreement* failure: a foreign expert clears a vision-only bar because its visual key collides with the query image. DEK therefore stores vision and text keys separately and requires both modalities to agree before an expert may fire.

A candidate i first passes the vision gate if it beats the prototype,

$$\mathrm{vision_ok}_i \;=\; \mathbb{I}\big[s_i^{\mathrm{v}} > s_{\mathrm{prot}}^{\mathrm{v}}\big]\,. \tag{3}$$

Among the current pool we also score prompt-key agreement. With margin $m \geq 0$ (often $m{=}0$ in our runs), the text gate keeps only experts near the best text match:

$$\mathrm{text_ok}_i \;=\; \mathbb{I}\left[s_i^{\mathrm{t}} \geq \max_j\, s_j^{\mathrm{t}} - m\right]. \tag{4}$$

Dual-Expert Keys retain expert i only under the conjunction

$$r_i \;=\; \mathrm{vision_ok}_i \;\wedge\; \mathrm{text_ok}_i. \tag{5}$$

Survivors are softly fused (softmax over gated vision logits) into a residual that is added at a chosen LLM layer; non-survivors contribute nothing. The dual-gate is inference-time hygiene: it does not rewrite f_θ, and it does not require the answer string to match y_i before rejecting a foreign expert. That is why FPR can fall sharply even when token-level image-swap remains only moderately improved.

Adversarial visual binding training

Dual-Expert Keys (DEK) assume that keys are at least somewhat grounded. Adversarial visual binding (AVBT) is the training counterpart: for each edit we optimize ordinary edit losses on the correct image, and we add an adversarial binding term on swap and hard-negative images built from the same prompt.

Let $\pi(\cdot)$ be the softmax routing distribution over pool experts plus the prototype slot, evaluated with binding-time vision scores. On the correct image x_i, with own-expert index set $\mathcal{O}_i$, the positive term encourages mass on the true expert(s):

$$\mathcal{L}_{\mathrm{pos}} = -\log\left(\sum_{i \in \mathcal{O}_i} \pi_i(p_i, x_i)\right). \tag{6}$$

On an adversarial image x^{adv} (swap or hard-negative) let π^{adv} be the same distribution, $m_{\mathrm{exp}} = \sum_{i \neq \mathrm{prot}} \pi_i^{\mathrm{adv}}$ the total expert mass, and $\pi_{\mathrm{prot}}^{\mathrm{adv}}$ the prototype probability. With hinge threshold $\tau \geq 0$ we penalize confident expert activation and push mass toward "no edit":

$$\mathcal{L}_{\mathrm{neg}} = \mathrm{ReLU}(m_{\mathrm{exp}} - \tau)\,\left(-\log \pi_{\mathrm{prot}}^{\mathrm{adv}}\right) + \mathcal{L}_{\mathrm{aux}}, \tag{7}$$

where $\mathcal{L}_{\mathrm{aux}}$ collects light auxiliaries used in our runs (a margin on the top wrong expert and an optional contrastive push). The adversarial visual binding objective is

$$\mathcal{L}_{\mathrm{bind}} = \mathcal{L}_{\mathrm{pos}} + \beta\, \mathcal{L}_{\mathrm{neg}}, \tag{8}$$

and the full training loss is $\mathcal{L} = \mathcal{L}_{\mathrm{edit}} + \lambda \mathcal{L}_{\mathrm{bind}}$. Unless noted otherwise we use $\lambda{=}0.3$, $\beta{=}2.5$, and $\tau{=}0.3$ with

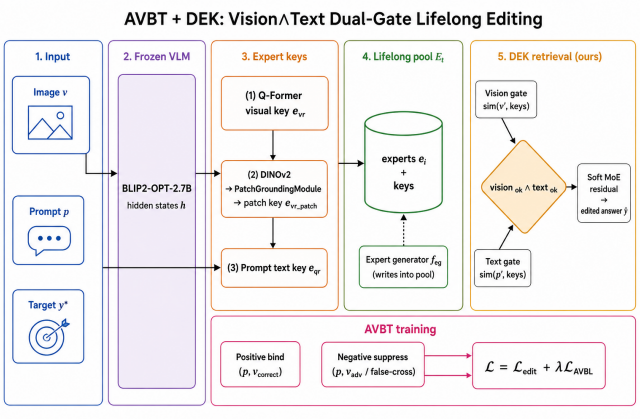


Figure 2: Overview of AVBT + Dual-Expert Keys on a frozen VLM. Each edit writes an MoE expert and separate vision/text keys into a lifelong pool. AVBT trains positive binding on the edited image and suppression on adversarial images. DEK retrieves only experts that pass both a vision gate and a text gate before soft MoE injection.

the contrastive auxiliary on. Binding training alone does *not* solve FPR on VLKEB (still $\approx 96\%$): it reshapes attachment, but a vision-heavy retriever can still admit foreign experts. That is why we treat Dual-Expert Keys as the primary anti-leakage mechanism and adversarial visual binding as supporting training.

Experiments

We want the evaluation to answer one question cleanly: under adversarial images, does Dual-Expert Keys reduce false-positive retrieval without giving up reliability on the edited photograph? Everything below is organized around that question rather than around chasing the highest token-swap score.

Backbone and editor testbed. All primary runs use BLIP-2 OPT-2.7B as a frozen vision-language backbone (Li et al. 2023). We edit it with a LiveEdit-style lifelong MoE editor (Chen et al. 2025): each training edit writes a low-rank residual expert into a growing pool, and inference retrieves which expert(s) to inject. We keep this architecture as the experimental testbed because continuous expert pools make wrong-expert retrieval observable; the scientific claim is about anti-leakage routing, not about beating LiveEdit on the original Rel, Gen or Loc leaderboard. Weight-updating multimodal editors (Fine-tuning, MEND-VL, SERAC-VL) (Mitchell et al. 2021, 2022; Cheng et al. 2023) appear as external comparisons. They can change answers, but they do not expose a growing expert identity that we can score for FPR, so FPR is reported only for pool-based systems.

Datasets and probes. Our main benchmark is the original VLKEB multimodal editing dataset (Huang et al. 2024). Each VLKEB example already provides an edit triple (p_i, x_i, y_i) together with the standard controls used in prior work: a text rephrase of p_i, an image rephrase of x_i, and text, multimodal locality items. We keep those fields unchanged. What VLKEB does not provide is an explicit test of whether retrieval still fires the right expert when the prompt is fixed and the image is wrong. We therefore build an adversarial evaluation overlay on top of the original VLKEB splits rather than introducing a separate benchmark.

Construction. Starting from the original VLKEB train and eval JSONs and the VLKEB image root, we attach two extra image probes to every retained edit:

- **Image-swap:** Keep prompt p_i and target y_i, but replace the image with x_j from another VLKEB edit $j \neq i$. Partners must come from a different entity folder in VLKEB's image layout, so the swapped photograph depicts a different scene or entity rather than a trivial crop of the same one.
- **Hard-negative:** Keep p_i and y_i, but replace the image with a confusable distractor that is also drawn from a different entity. We prefer partners whose prompts share a coarse question skeleton with p_i (same leading tokens), which makes the distractor harder than a random unrelated photograph.

If a candidate image file is missing on disk we skip that partner and try the next-ranked one; edits without a valid swap and hard-negative pair are dropped. The resulting files still look like VLKEB rows: they retain Rel , Gen and Loc fields and simply add the two adversarial image paths used at evaluation (and, for binding training, at training). Reliability, generality, and locality are therefore measured on the same VLKEB content as usual; only the adversarial probes are ours.

Training vs. evaluation use. The lifelong baseline is trained on the original VLKEB training edits. Adversarial visual binding and Dual-Expert Keys additionally consume the adversarial images from the probe-augmented training split, because the binding loss needs swap , hard-negative views of each edit. All primary tables evaluate the same probe-augmented VLKEB evaluation split under a frozen editor pool. As a cross-dataset check we apply the identical pairing procedure to EasyEdit EVQA examples (Wang et al. 2024), again without changing the editor itself.

Method	Rel↑	T-Gen↑	M-Gen↑	T-Loc↑	M-Loc↑	FPR↓
Lifelong baseline	93.5	94.3	92.1	100	100	96
Binding only	93.8	94.5	93.8	100	100	96
Dual-Expert Keys (ours)	93.7	94.5	92.0	100	100	**46**

Table 1: Main VLKEB result. Standard metrics stay on parity with the lifelong baseline; Dual-Expert Keys cut false-positive retrieval $96\% \rightarrow 46\%$.

The probe suite used throughout is therefore: (i) reliability on the edited image x_i; (ii) image-swap; (iii) hard-negative; (iv) VLKEB text , image rephrase generality and locality (Huang et al. 2024; Chen et al. 2025).

Metrics. Standard multimodal editing metrics remain in the tables for context: reliability (Rel), text generality (T-Gen), multimodal generality (M-Gen), and text , multimodal locality (T-Loc , M-Loc) (Cheng et al. 2023; Huang et al. 2024). On adversarial probes we also report answer-level image-swap and hard-negative accuracy, plus the reliability-swap gap. Our primary novelty metric is **false-positive retrieval** (FPR): the fraction of adversarial queries for which routing activates at least one expert other than the intended edit (lower is better). FPR is computed from routing logs under the same evaluation harness as Rel, swap; it is deliberately a retrieval-safety number, not a substitute for token accuracy. When tables include both answer-level and routing metrics, we read them in that order: FPR first for the leakage claim, then Rel to confirm the edit still works, then swap, hard-negative and gap as secondary behaviour.

Baselines, comparisons, and ablations We compare five rows that isolate mechanism:

- **Lifelong baseline:** LiveEdit-style MoE retrieval without adversarial visual binding and without Dual-Expert Keys.
- **Binding only:** adversarial visual binding training with patch grounding, still using the original retrieval rule.
- **Dual-Expert Keys (ours):** adversarial visual binding plus the vision $\wedge$ text retrieval gate.
- **Neighbor mining , pool buffer:** Dual-Expert Keys plus false-cross neighbor mining, or Dual-Expert Keys plus a larger pool buffer reported only as negative and secondary ablations.
- **Fine-tuning, MEND-VL and SERAC-VL:** weight-updating multimodal editors under the same VLKEB probes.

In primary runs adversarial visual binding uses λ=0.3, β=2.5, hinge τ=0.3, and the contrastive auxiliary described above. Dual-Expert Keys applies the vision $\wedge$ text conjunction gate; the text margin m is held fixed across primary runs.

Evaluation protocol For the main VLKEB tables we evaluate 100 edits with 10 generations per example. The headline ablation and competitor numbers use seed 42; multi-seed credibility uses five evaluation seeds $\{42, 0, 1, 2, 3\}$ on the same checkpoints. EVQA stress follows the same three-way contrast lifelong baseline, binding only, and Dual-Expert Keys with a larger multi-seed subsample of 300 edits when we report aggregate means and standard deviations. Throughout, we log routing identities whenever false-positive retrieval is claimed. We do *not* treat token-level image-swap as a submission gate: the intended claim is a reduction in false-positive retrieval with reliability held.

Results

Unless noted otherwise, all VLKEB numbers use the probe-augmented original evaluation split (100 edits, 10 generations per example). We lead with false-positive retrieval (FPR; lower is better) at reliability parity: that is the claim Dual-Expert Keys are designed to win. Token-level image-swap is reported as secondary behaviour, not as a submission gate.

False-positive retrieval on VLKEB

Table 1 is the headline operating point. Dual-Expert Keys preserve LiveEdit-style reliability, text,image generality, and locality, while cutting FPR from 96% to 46% a 50 percentage-point absolute reduction. Binding training alone leaves FPR unchanged at 96%: it can move answer scores on adversarial images, but it does not fix wrong-expert retrieval. The dual vision-text gate is what changes the retrieval decision.

What each component contributes

Table 2 isolate the stack. Moving from the lifelong baseline to binding-only training raises image-swap accuracy ($40.7\% \rightarrow 47.8\%$) but leaves FPR at 96%. Adding Dual-Expert Keys then collapses FPR to 46% while reliability stays near 94% and swap stays near the binding-only level. Extra neighbor mining trims FPR only slightly ($46\% \rightarrow 43\%$) without a swap gain; enlarging the pool buffer *raises* FPR again to 55%. We therefore treat neighbor mining and the pool buffer as negative ablations, not as part of the method row.

Method	Rel↑	Swap↑	FPR↓	Role
Lifelong baseline	93.5	40.7	96	baseline
Binding only	93.8	47.8	96	+binding
Dual-Expert Keys	93.7	47.7	**46**	+dual gate
+ Neighbor mining	93.7	47.2	43	negative
+ Pool buffer	93.7	48.0	55	negative

Table 2: Ablation of the lifelong editor family on VLKEB (seed 42). Dual-Expert Keys are the FPR mechanism; binding alone is not.

Multi-seed stability and significance

Figure 3 and Table 3 repeat the lifelong-baseline vs. Dual-Expert Keys comparison over five evaluation seeds on the same checkpoints. FPR is 96% for the baseline and 46% for Dual-Expert Keys on *every* seed (−50 pp, no regression). Mean image-swap rises from 41.2% to 48.6% (+7.4 pp; paired t-test, n=5, two-sided p=3.4×10^{-4}). Reliability remains held (93.5% vs. 93.7%). The FPR win is therefore not a single-seed artefact: the dual-gate decides retrieval in a stable way, and the modest swap lift is statistically significant.

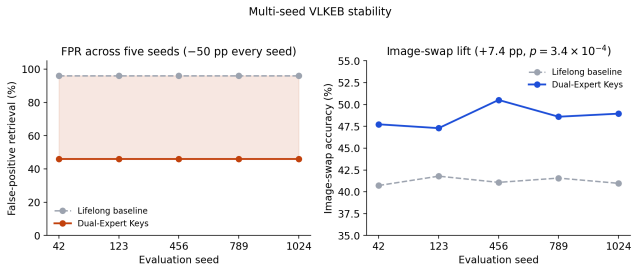


Figure 3: Five-seed VLKEB stability. Left: false-positive retrieval stays at 46% for Dual-Expert Keys versus 96% for the lifelong baseline on every seed. Right: image-swap accuracy improves on every seed (+7.4 pp mean).

Method	Rel↑	Swap↑	FPR↓	Swap-gap↓
Lifelong baseline	93.5	41.2±0.4	96.0	52.2±0.4
Dual-Expert Keys	93.7	**48.6±1.3**	**46.0**	45.1±1.3

Table 3: Multi-seed VLKEB summary (mean ± std over five seeds; Rel/FPR constant across seeds here). Swap lift +7.4 pp (p=3.4×10^{-4}); FPR −50 pp.

Comparison to weight-updating editors

Table 4 place Dual-Expert Keys next to Fine-tuning, MEND-VL, and SERAC-VL under the same VLKEB probes. The external editors either under-edit or collapse reliability: Fine-tuning and MEND-VL sit near 71-73% Rel with almost no reliability-swap gap, and SERAC-VL falls to ≈ 39% Rel. None of them expose a growing expert identity, so FPR is not defined for those rows. Among systems that *do* maintain a lifelong pool, Dual-Expert Keys are the only setting that keeps Rel near 94% while cutting FPR in half. That is the intended contrast: not a race for the highest adversarial token score, but a high-reliability anti-leakage operating point.

Method	Rel↑	Swap↑	Swap-gap↓	FPR↓
Lifelong baseline	93.5	40.7	52.8	96
Binding only	93.8	47.8	46.0	96
Dual-Expert Keys (ours)	93.7	47.7	46.0	**46**
Fine-tuning	73.0	73.1	≈0	n/a
MEND-VL	71.3	69.4	1.9	n/a
SERAC-VL	38.6	38.6	≈0	n/a

Table 4: Competitors on VLKEB probes (seed 42). FPR applies only to retrieval-pool editors.

Cross-dataset stress on EVQA

To check that the FPR story is not VLKEB-specific, we apply the same probe construction and the same lifelong-baseline , binding-only and Dual-Expert Keys contrast to EasyEdit EVQA (Wang et al. 2024). Table 5 summarize five evaluation seeds with 300 edits each. Reliability stays near 95% for all three rows. Binding only again tracks the lifelong baseline on FPR. Dual-Expert Keys reduce image-swap FPR to 1.5%±1.1% and hard-negative FPR to 2.7%±1.3% more than an order of magnitude below the baseline hard-negative leakage (96.8%±0.6%). Token-level swap on EVQA remains low for every method; we do not claim an EVQA swap leaderboard. The transferable message is the same as on VLKEB: Dual-Expert Keys suppress wrong-expert retrieval while holding reliability.

Method	Rel↑	FPR_{swap} ↓	FPR_{hard} ↓
Lifelong baseline	95.0±0.6	31.3±38.1	96.8±0.6
Binding only	95.2±0.6	30.3±38.8	97.0±0.4
Dual-Expert Keys	**95.2±0.5**	**1.5±1.1**	**2.7±1.3**

Table 5: EVQA multi-seed grounding stress (300 edits per seed). Means ± std over seeds $\{42, 0, 1, 2, 3\}$.

Limitations

Our primary claim is retrieval safety, not token-level image-swap accuracy; residual swap answer headroom is a separate generation problem for the frozen decoder. False-positive retrieval is also defined only for lifelong editors with an identifiable expert pool, so FPR is n/a for weight-updating baselines such as Fine-tuning, MEND-VL, and SERAC-VL. The main evidence uses BLIP-2 OPT-2.7B on VLKEB, with EVQA as a completed cross-dataset check under the same dual-gate rule (Li et al. 2023; Huang et al. 2024; Wang et al. 2024).

Conclusion

Lifelong multimodal editors can look correct on the image they were edited for and still retrieve the wrong stored fact when the scene changes. We identify that failure as *false-positive retrieval* and show that standard metric scores do not catch it. Dual-Expert Keys fix the decision rule: an expert may fire only when vision and text agree. Adversarial visual binding strengthens edit-image attachment during training so the dual gate is not fighting weakly grounded keys.

These mechanisms translate into large, stable reductions in false-positive retrieval. On original VLKEB with our adversarial probes, Dual-Expert Keys reduce false-positive retrieval from 96% to 46% while keeping reliability near 94%. The cut is stable across five evaluation seeds, survives comparison against Fine-tuning, MEND-VL, and SERAC-VL at a high-reliability operating point, and reappears under EVQA stress. Binding training alone does not deliver the FPR win; the dual gate does. Taken together, these results show that lifelong multimodal editing can be made substantially safer at retrieval time without sacrificing the reliability that existing benchmarks already reward.

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